

ActInf GuestStream #038.1 ~ Max Berg "Oversampled and undersolved"

GuestStream | Max Berg | 1:31:05

Hello and welcome. It's March 9th, 2023. We're an active guest stream number 38.1 here with Max Berg.

Thank you, Max, for joining. This is going to be great. Really looking forward to your presentation

and discussion. So appreciation again for joining and looking forward to what you'll share with us

today. Yeah, thank you, Daniel, for having me. As I was introduced, my name is Max Berg. I come from

the University of Marburg and I'm a clinical psychologist. I want to do a little bit of

impression management first. So Marburg is next to Frankfurt. In case you don't know, it's a

small or medium city in Germany. And regarding me, I'm interested in the intersection of clinical

psychology, neuroscience and philosophy. That doesn't mean that I'm an expert in all of these

fields. Of course, I have a lot of knowledge in clinical psychology, I would say, and the rest

is more of my hobbies. And so it's important that you keep in mind to expect a clinical perspective

today. I'm not an expert on like Bayesian mathematics or anything like that. I know a little bit about it

and I've read books about it and try to keep myself updated, but it's not my expertise. My expertise is

clinical psychology. And what I want to present today is like a paper about depressive rumination.

But we will see that it's not important that it's depressive rumination and rumination is a much

broader construct here. And we want to rephrase like depressive rumination from an active inference

perspective. And this is what we want to talk about today. Yeah. So I want to start with a little quote

and it's rumination is a coin of my realm. Interiority breeds interiority. And we will see that this might

be literally true. And this is why I think it's a good quote to start with. So what is rumination?

It is defined by the APA as obsessional thinking involving excessive repetitive thoughts or themes

that interfere with other forms of mental activity. It is a common feature of OCD and generalized anxiety

disorder. I actually don't know why these two disorders were chosen first, but well, here we go. I just

quoted them here. I would add major depression to the list. And at least in my mind, major depression is

the prototype disorder for rumination. But sure, OCD and generalized anxiety disorder also has a lot of

rumination going on. And here you can see a first really important point, namely that rumination is a

transdiagnostic marker, right? It's important for multiple psychiatric conditions.

And yeah, as said, it's an important transdiagnostic risk factor. And it can predict actually depression

and scores and subsequent measurement time points. And this is what makes it interesting. There was like a

grant, a small grant proposal by, I believe, the Wellcome Trust in the UK, where they gave grants for

writing systematic reviews about transdiagnostic topics in mental health. And unsurprisingly,

rumination is one of the topics where they gave or where they wanted to have systematic reviews for

which, yeah, they can see that it's an important topic where people really believe that it has some

has some utility, clinical utility. And unfortunately, it is also the case that it's associated with worse

treatment response for psychotherapy and also for medication treatment.

Yeah. And let's make it more complicated, shall we? The rumination is sometimes used interchangeably with

the concept of repetitive negative thinking. And sometimes repetitive negative thinking means rumination

and varying. But we will use the definition of rumination in its broader sense today. So when I talk about rumination, I also mean varying some of the time. I just wanted to make this clear at the beginning.

So I use it in the older broader sense, so to say.

Yes. So why would we need a new concept for rumination? Well, reinforcement learning, I would say, struggles to explain the constant continuation of ruminative thinking. To make it short, why should you continue with doing something that is not giving you like pleasure or any good feelings in the short or the long run? And yeah, why is something continued despite not promoting action and also leading to depressed mood?

This is a question at hand. It's not an easy question to answer, I would say. And this is why we would say that we have to not only take exploitation learning into account, but also take exploration learning or surprise reducing learning or free energy minimization into account, however you want to call it. The important

part is you have to take into account that not all learning is done in the sense of like reinforcement learning, but there's also learning that we do to know more about the world. And if this goes wrong, rumination can occur. And this is how we want to rephrase it today. Yeah. So this is the roadmap for today. We want to reconceptualize rumination as an extreme case of failed mental problem solving. We want to present a model of rumination using active inference terminology. So this will be a conceptual model. We don't have the mathematics to back it up at the moment. I will talk about this limitation at the end again. And we will also discuss neurobiological correlates shortly to then continue to point towards treatment implications of the model. So the next part is like, I want to talk a little bit about problem solving and mental problem solving. And I've had a classic quote by Richard Dawkins that I really find interesting in this context. It says, survival machines that can simulate the future are one step ahead of survival machines who can only learn on the basis of overtrial and error.

The trouble with overtrial is that it takes time and energy. The trouble with overt error is that it's often fatal. Simulation is both safer and faster. Yeah. And I think this is really interesting. And we can talk about also like this evolutionary perspective later. I have really broke it down into simple terms first. Okay. So when I want to solve problems using my mind, what do I need to do? Well, at first, I need to think about what to do next. What do I want to think about, so to say, the... And we have called this step candidate sampling. So it's meter policy basically. So what do I even want to think about before thinking about

what are important tasks of today, for example?

And then the next step is policy evaluation. Or if I think about that, what are the expected results? If I tell them about this concept, will they understand my talk today, for example? And then let's say, yes, I hope that this is the case.

And then we have to do another step and do like the actual problem solving. Okay. Now I'm convinced that this might

be the right way, but how to do it, how to actually solve the problems, which are the steps necessary.

And these three steps we will consider over and over again today. And we have in the paper a neat little example that happens to me a lot. That's one of the reasons I re-chose it today. Imagine you lost your key.

And now what we typically do is at first we check the usual places. Oh, have I put it there? No. Have I maybe put it there? No.

No. And after a while, if this doesn't work, my policy changes typically. And I think, okay, um, where did I see my key the last time that I saw it and trace it back? And maybe this works and I find my key back. And a few days after I have the same problem again, because I already lost my key again. And what I may do after a few instances of this behavior is like, okay, I need to change something fundamentally in order to get my key back faster. And so in Bayesian learning terms, at the beginning, my expected prediction error minimization is just search the usual places. It might work. Okay. After a while, the mental, uh, the mental policy, okay, why do I keep losing the key? And how do I solve this problem? Might lead to a higher expected free energy minimization or prediction error reducing properties. And this is why I switched the policy right after a while. So this meter policy step decides how to go or how to go about a given problem. In this case, how to find my key.

Michael Hanna, Ph.D.: Good. This is really important. We cannot search the interactively large probability risk distribution for this step. It's not possible to remember all instances where I lost my key and to compare like promising strategies based on the whole distribution. I'm not able to do this. And

Michael Hanna, Ph.D.: So if no machine, no living being, we have to find fitting samples, so to say, and fitting means

Michael Hanna, Ph.D.: It's usually based on heuristics. What did I do the last time when the key was gone? How can I find it? This will become important later because these heuristics can run into problems. They help us

Michael Hanna, Ph.D.: Like make computationally intractable problems solvable, but these heuristics have also their downside as we're going to discuss.

Michael Hanna, Ph.D.: Okay, then step two. Now policies are then selected if they provide a trade-off between pragmatic and epistemic value.

Michael Hanna, Ph.D.: And pragmatic is okay. They I have a good chance of finding my key back to stay in the example and epistemic is I learn also something about finding keys or understand the world better.

Michael Hanna, Ph.D.: And this is also important whenever I choose a specific policy, high opportunity costs may arise. What I mean with that is like when I pursue like a searching process, for example,

Michael Hanna, Ph.D.: For finding my key, I cannot continue preparing my talk or something like that. So every decision I make and has necessary costs because I cannot pursue other goals in the same time.

Michael Hanna, Ph.D.: And this is also becoming going to become important. So in order to be chosen by an organism, each chosen policy has to outweigh the resulting opportunity costs.

Michael Hanna, Ph.D.: Otherwise, it's not adaptive.

Michael Hanna, Ph.D.: Right. Okay, and now the last step.

Michael Hanna, Ph.D.: The mental problem solving good. I now decided for one strategy of finding my key.

Michael Hanna, Ph.D.: And now I have to actually pull it off. So where to search what to do next, which sequences of movements do I have to perform

Michael Hanna, Ph.D.: And how to go about it best and typically in different strands of literature, not just in active inference, I need to perform something that is called a tree search, you know, this from chess playing typically or chess computers.

Michael Hanna, Ph.D.: Like they do it a lot better than we do, but they look okay if I do this move and he does this move, then I can do this move and in all that follow it, I have a good position.

Michael Hanna, Ph.D.: And it's again not possible to search the whole tree because the more branches it becomes,

Michael Hanna, Ph.D.: The more branches are daughter branches of these branches again, so the tree grows exponentially.

Michael Hanna, Ph.D.: So I need to have a strategy of not searching through all the branches. It's called a pruning strategy in the case of Bayesian learning is often Occam's window, meaning.

Michael Hanna, Ph.D.: That branches are not searched through if they don't promise a prediction error minimization that is sufficient to pursue searching them.

Michael Hanna, Ph.D.: This can also lead into problems because if a branch at first doesn't provide enough prediction error minimization it will not be searched through despite the fact that it could have some prediction error minimization further down the road, so to say.

Michael Hanna, Ph.D.: But we have to prune again because it's not possible to search the whole distribution.

Michael Hanna, Ph.D.: Okay.

Michael Hanna, Ph.D.: So with these three steps in mind, we can have the following model. So we have the candidate sampling,

Michael Hanna, Ph.D.: how to go about the lost key, so to say policy evaluation. If I do the strategy, have I do I have a good chance of finding the key and learning something about the world and mental problem.

Michael Hanna, Ph.D.: How to actually pull the strategy of and I want to say one really important thing, and this is an increase in mental problem solving is a normal process in a lot of situations. Right. So,

Michael Hanna, Ph.D.: For example, if I lose a person that is really important for me it's normal that my general model set decreases like a lot of my strategies for getting social supports don't work anymore because the person's gone and so

Michael Hanna, Ph.D.: In in in in Bayesian terminologies prediction errors increase after such an event and this is a normal process.

Michael Hanna, Ph.D.: And if like my general model declines this was linked in the active inference literature to bad mood or to feelings of sadness, for example,

Michael Hanna, Ph.D.: And in such cases typically you think more about fundamental things in your life. Everybody knows it and has experienced it. If you don't feel well, you tend to think more and to try to get into a better situation again.

Michael Hanna, Ph.D.: And hopefully what happens then is okay I

Michael Hanna, Ph.D.: For example, I lost someone my social support system is breaking together.

Michael Hanna, Ph.D.: Or is breaking apart falling apart. So after a while I search for people that I like and that can form a new bond with.

Michael Hanna, Ph.D.: And hopefully after a while my prediction errors become smaller again, but therefore I need to take action. I need to think about what to do with my life and how to get to know new people in this example to like reduce my.

Michael Hanna, Ph.D.: prediction errors in this case regarding social support and then I have to pull it up. I have to go to certain places to meet people.

Michael Hanna, Ph.D.: And hopefully after a while of sadness and difficult times.

Michael Hanna, Ph.D.: My general model fit increases again.

Michael Hanna, Ph.D.: But this is not always the case as we now want to discuss next. So I have yet another quote you will get to you get to use two quotes today. Sorry.

Michael Hanna, Ph.D.: The next quarter is about rumination and assess shut up. She tells her monkey mind. Please shut up. You pick off nits press off bruises counter of losses zero failures collector of grievances future and past.

Michael Hanna, Ph.D.: And now we want to discuss what happens when this doesn't work. So it can happen that despite trying agents and organisms don't find good approximations for important problems in their life right.

Michael Hanna, Ph.D.: And if this is the case, the prediction errors regarding these problems stay high because they surprise doesn't get resolved and

Michael Hanna, Ph.D.: Despite trying the organism tries over and over again prediction errors say hi surprise doesn't get resolved.

Michael Hanna, Ph.D.: And the prediction error doesn't reduce again and the problem stays and what we now want to do is discuss reasons for why this happens in depressive rumination.

Michael Hanna, Ph.D.: And at first we have to say, okay, a typical rumination problems have a high complexity with uncertainties at many hierarchical levels. If you think about problems, for example, like

Michael Hanna, Ph.D.: Why is the world such a cruel place, for example, this could be a nice rumination problem.

Michael Hanna, Ph.D.: You would have to gather a lot of evidence from different strains. It's really hard to actually disprove such a claim or to prove such a claim or even to say what it means.

Michael Hanna, Ph.D.: And this is why there might be no efficient solution for such a question. It's not like navigating a maze or a maze or a maze or a maze or a maze or a maze.

Michael Hanna, Ph.D.: But it's much more complicated. And in addition, sometimes like we human beings have very high precision demands for solutions. For example, after I may have survived a traumatic event, I might be really reluctant to go to similar looking places again.

Michael Hanna, Ph.D.: And I have to be really, really sure that the new place.

Michael Hanna, Ph.D.: Where I don't want to go is really different from the place where something bad happened to me. And if I'm not sure and sure means really sure high precision. I won't go there.

Michael Hanna, Ph.D.: And

Michael Hanna, Ph.D.: And

Michael Hanna, Ph.D.: These complexity demands can also make an efficient approximation to a problem improbable.

Michael Hanna, Ph.D.: And

Michael Hanna, Ph.D.: This might be one reason why problem solving isn't functioning in the case of depressive rumination.

Michael Hanna, Ph.D.: Another reason might be over pruning and missing action policies. So as I said beforehand, like pruning in order to

Michael Hanna, Ph.D.: function properly.

Michael Hanna, Ph.D.: Or let's say this way pruning is a good strategy when the

Michael Hanna, Ph.D.: Parts of the search tree that I remove, so to say, don't yield like at the beginning small losses and at the end large gains because if it's like that the tree branch might be pruned away despite.

Michael Hanna, Ph.D.: Actually helping the organism to reduce surprise, but the organism will never find out because it gets pruned away.

Michael Hanna, Ph.D.: And the next thing is also action policies can all only be thought through if an agent knows about them, you know about a lot of depression patients with a lot of childhood adversities that actually don't know a lot of strategies for navigating social environments.

Michael Hanna, Ph.D.: It's not their fault. It's the consequence of childhood adversities, but nevertheless, this is a problem. If I don't know the solution for a social problem, I cannot approximate a solution because action policies are missing.

Michael Hanna, Ph.D.: And the next problem might be that after many failed attempts of problem solving, I might not even have a reason to solve it.

Michael Hanna, Ph.D.: Engage in deeper instrumental learning what I mean with that I go back one slide.

Michael Hanna, Ph.D.: After a while I have an idea. So, okay, why is my life becoming so shitty.

Michael Hanna, Ph.D.: It is because I don't engage with other people anymore. I just lie in my bed and feel sad. Okay.

Michael Hanna, Ph.D.: And now normally there would be a step where I think about how to engage with other people again. But if I have, for example, missing action policies really bad mood and not a lot of strategies.

Michael Hanna, Ph.D.: I might think this through shortly and think, well, I can try, but they won't like me anyway. So instead of deep mental problem solving, I jump back to step one, which is why do I have so many problems with other people.

Michael Hanna, Ph.D.: And there you can see a ruminative circle is forming and no deeper engagement is possible anymore.

Michael Hanna, Ph.D.: And if this happens, my prediction errors stay large and more candidate sampling occurs. Okay.

Michael Hanna, Ph.D.: I cannot solve this by meeting other people, but maybe I can do it on my own. No, I feel so sad and lonely. You see where this is going. I do not engage anymore in deeper problem solving, but instead thinking through a lot of policy candidates and none of them is like giving me confidence that I can solve my problems.

Michael Hanna, Ph.D.: So I stay at the same meter policy selection steps, so to say, and in consequence, I have less instrumental thinking and less instrumental learning.

Michael Hanna, Ph.D.: And this also has consequences.

Michael Hanna, Ph.D.: For example, the pragmatic value of policies are estimated to be low due to low model fit and this even becomes worse over time.

Michael Hanna, Ph.D.: So if I think I cannot do it, I will not train it. If I do not train it, I become even worse at doing it. And so I won't, I will, it becomes more and more improbable that I try because my action policies become more and more under trained.

Michael Hanna, Ph.D.: And yeah, this is another really important problem.

Michael Hanna, Ph.D.: And the next problem is the opportunity costs because I ruminate a lot. So now we are at a meter level. I don't do much else. And because I don't do much else, my model fit further declines. And yeah, the rest we talked about on the last slide. So.

Michael Hanna, Ph.D.: Ruminative states may be learned as the agent's default response for quote unquote dealing with problems after a while and now we have a vicious cycle here.

Michael Hanna, Ph.D.: And the next thing that I want to discuss is how neurobiology might play a role here and I have had a little joke of Phil Collette ripped off for my slides here.

Michael Hanna, Ph.D.: And quotation by Carl Friston that says an agent does not have a model of its world. It is a model in other words, the form structure and states of our embodied brains do not contain a model of the sensoriums. They are that model.

Michael Hanna, Ph.D.: And on the right, you can see like Yale School of Management guy that actually really embodies this principle as he's clearly showing on his backpack.

Michael Hanna, Ph.D.: Okay, jokes aside.

Michael Hanna, Ph.D.: What is first to be said, the default mode network is going to be important for our discussion here. The default mode network is a large scale network with medial frontal temporal and

Michael Hanna, Ph.D.: regions in the brain. It's important for introspection self referential thinking and autobiographical recall. So in short, it's important for everything mental problem solving and rumination related or a lot of these things and this is why it's unsurprisingly going to play a role here.

Michael Hanna, Ph.D.: And

Michael Hanna, Ph.D.: To be more precise, the frontal cortices seem to be the main driver of memory recall in the default mode network. So if the default mode network tries now I'm intramorphizing it and to recall a memory. So

Michael Hanna, Ph.D.: Where do I find the key, for example, it uses the frontal regions to be the main driver of this process and

Michael Hanna, Ph.D.: It is the case that meta studies found a stronger coupling and a hyper activation in default mode network structures to be correlated with with ruminations and they're actually two meta analysis that found this

Michael Hanna, Ph.D.: And it is also suggested that a coupling with subcortical cortex might contribute to the negatively valence nature of rumination MDD.

Michael Hanna, Ph.D.: So we know okay DMN is important for problem solving and for rumination and it's also found meta analytically that it indeed plays a role for rumination or at least it's correlated.

Michael Hanna, Ph.D.: And

Michael Hanna, Ph.D.: In addition, meta studies also found evidence for a reduced frontal parietal control network within network connectivity in major depression.

Michael Hanna, Ph.D.: And this is also interesting because as we discussed our idea is basically that rumination means I get stuck in abstract thinking.

Michael Hanna, Ph.D.: And don't use my instrumental learning capabilities as much, and so it makes sense that in MDD we find reduced control network or frontal parietal control network activity.

Michael Hanna, Ph.D.: Note, by the way, that this part of the evidence line is more indirect than the part before right I don't want to gloss over this so.

Michael Hanna, Ph.D.: Why we have here a direct correlation with rumination in the case of the default mode network here it's more MDD in general, this is why we also wrote it much more careful in the paper.

Michael Hanna, Ph.D.: Because it's a more indirect line and I don't want to gloss over that and.

Michael Hanna, Ph.D.: It is further more known that a connection between control networks and default mode

networks is seem to be reduced and interestingly, these two networks have to.

Michael Hanna, Ph.D.: connect in order to solve complicated problems they actually study.

Michael Hanna, Ph.D.: about that and the study suggests that if the problem gets harder for an individual it uses more of the default mode network to do the meter thinking and then the control networks to do the problem solving and the interplay between the two is really important.

Michael Hanna, Ph.D.: And okay so in conclusion and what we think could be the case is that the default network is hyperactive and hyper connected because it does the policy candidate over sampling that we talked about before.

Michael Hanna, Ph.D.: And the reduced mental problem solving or instrumental learning and due to the policy over sampling is reflected by the reduced resting state activation and the hyper connectivity of the control networks and also of integration hubs like the insular.

Michael Hanna, Ph.D.: Okay.

Michael Hanna, Ph.D.: Okay.

Michael Hanna, Ph.D.: Then a short citation, it will be the last one and it's an existentialist model and it says simply existence precedes essence and it's wants us to say okay.

Michael Hanna, Ph.D.: In the first instance, we are beings that get thrown into the world, but we have to do something with our lives in order to flourish and give it give ourselves.

Michael Hanna, Ph.D.: reasons to live and to succeed and I want to talk about a little bit about like how can we get out of ruminative loops and what might be old idea and new ideas.

Michael Hanna, Ph.D.: At first, I want to say there's good evidence for the effectiveness of extreme of existing psychotherapeutic treatment regarding depression in general.

Michael Hanna, Ph.D.: And it's also regarding rumination so we also have like meter analysis specifically analyzing rumination, however.

Michael Hanna, Ph.D.: We have substantial non responder rates for psychotherapy and also for current medication approaches.

Michael Hanna, Ph.D.: We have a high chance for relapse after depression is gone, it will oftentimes reoccur in the life of patients, unfortunately.

Michael Hanna, Ph.D.: And we also have substantial therapy dropouts so patients actually leaving therapy.

Michael Hanna, Ph.D.: So it's really easy to say that psychological and pharmacological treatments needs improvement.

Michael Hanna, Ph.D.: So they are effective, but they don't help everyone and there's a lot of room for improvement here.

Michael Hanna, Ph.D.: And there are multiple ideas that we briefly discussed in the paper and the first is like a simple idea, but could be quite effective and has also meter analytical evidence that it helps.

Michael Hanna, Ph.D.: And this is like reduce memory interference and rumination by physical exercise in the everyday life of patients.

Michael Hanna, Ph.D.: So doing more like especially endurance exercises actually helps reducing rumination, even when compared to an active treatment like CBT.

Michael Hanna, Ph.D.: So maybe depression patients don't only need therapy to be paid for, but also to have a personal trainer that might be paid for.

Michael Hanna, Ph.D.: And I only say this half jokingly, this could improve the life of a lot of people.

Michael Hanna, Ph.D.: And another idea is more medical it's.

Michael Hanna, Ph.D.: Psilocybin or ketamine was were shown in studies to increase the sampling entropy of the brain, and this could also be an interesting prospect, because we've learned that the samples are getting repeated over and over again, but.

Michael Hanna, Ph.D.: If we could have an idea for how to increase the variance of samples drawn then probability that one sample is actually a good idea for thinking through increases.

Michael Hanna, Ph.D.: However, I have to say cautionary notes, especially with.

Michael Hanna, Ph.D.: Psychedelic drugs because.

Michael Hanna, Ph.D.: It is a promising pharmacological avenue, probably, but at the moment we have really small sample sizes.

Michael Hanna, Ph.D.: It's hard to blind for these drugs between groups, I mean you know when you're tripping it's quite easy to know.

Michael Hanna, Ph.D.: So this makes RCT is difficult and and we know from a lot of questionable research practices from this line of research, unfortunately, I don't want to say that everybody in these fields are doing bad science it's not what I want to say, but.

Michael Hanna, Ph.D.: I know from a few cases where pre pre registered protocols don't fit the results that were presented and stuff like this and.

Michael Hanna, Ph.D.: If you combine all this it's actually quite hard to say whether these work really well, so what we would need is really larger study.

Michael Hanna, Ph.D.: Not the small brain imaging studies with 20 subjects more like 200 subject would be a beginning.

Michael Hanna, Ph.D.: And we need.

Michael Hanna, Ph.D.: The researchers to actually stick to predefined interpretations of the results, so we can learn about more about this problem promising avenue okay and more ideas would be.

Michael Hanna, Ph.D.: To change therapy a little bit and concrete idea is.

Michael Hanna, Ph.D.: Could actually.

Michael Hanna, Ph.D.: Add problem complexity estimations into therapy, what I mean with this is like ask patients okay how much time you spend ruminating.

Michael Hanna, Ph.D.: What typically are the results of these processes.

Michael Hanna, Ph.D.: What are the costs of this processes so we could try to make the costs of the process more explicit to the patient.

Michael Hanna, Ph.D.: And to give them a grasp on how much opportunity costs they invoke on themselves by the process it's not typically done at the moment, but it could be an idea for improving CBT.

Michael Hanna, Ph.D.: And.

Michael Hanna, Ph.D.: What is also important, this is done in CBT what I know tell by the way, but it's really important to keep it in mind.

Michael Hanna, Ph.D.: We learned that we good problem solving requires an intricate interplay between default mode network and brains control networks.

Michael Hanna, Ph.D.: So it's actually important to have optimal difficulties for behavioral tasks given to patients if it's too difficult, they will not do it.

Michael Hanna, Ph.D.: And if it's too easy, they will not learn anything from that no epistemic value is mind so to say, and both options are not good for reducing rumination.

Michael Hanna, Ph.D.: And there's actually another experimental treatment that shows some success and it's called emotional flanker tasks and in these tasks.

Michael Hanna, Ph.D.: Like patients learn how to shift their attention away from bad emotions and this could also help like stop the ruminative process because we learn.

Michael Hanna, Ph.D.: That depending on your mood you will have different heuristics in your brain about what to think about next and in the case of bad moods you don't have the best heuristics so this is another idea for a treat for better treatment.

Michael Hanna, Ph.D.: Yes.

Michael Hanna, Ph.D.: Now I want to talk about a few limitations and then we can enter the discussion so one limitation is clearly that the model only discusses typical depression where behavior breaks down.

Michael Hanna, Ph.D.: However, there are cases of agitated depression where behavioral output actually increases or to make it more abstract depression is a really heterogeneous subject and also rumination is really heterogeneous.

Michael Hanna, Ph.D.: So we don't have a model actually at the moment that considers all individual trajectories.

Michael Hanna, Ph.D.: So this is only a general scheme, so to say, but it will not fit every individual.

Michael Hanna, Ph.D.: And the next thing is we describe plausible ways for rumination development, but we do not know whether these pathways play a causal role in MDD development.

Michael Hanna, Ph.D.: I said briefly that when I have rumination at time point T , I will have more depression at time point T plus one, but we do not know whether this is a causal factor, but.

Michael Hanna, Ph.D.: Or only a coincidence in time, you know what I mean, so this is also an unsolved problem at the moment and.

Michael Hanna, Ph.D.: The last problem, which is also important for the active influence not see in the Community is it's not a mathematical model.

Michael Hanna, Ph.D.: It's a conceptual model that we did and so the next step would be to make a mathematical model out of it, for example, like partially observable mark of decision process or something like this.

Michael Hanna, Ph.D.: And in order to be able to simulate it and to check whether the behavior that we said it would come out actually come out, but this is another major limitation of the paper.

Michael Hanna, Ph.D.: Yes, I will skip over that I guess.

Michael Hanna, Ph.D.: We talked about this enough.

Michael Hanna, Ph.D.: So what I want to talk about shortly before entering the discussion is other projects that we want to do next with respect to active inference and.

Michael Hanna, Ph.D.: What we actually have done and now it under review at psychological medicine is applying the active inference framework to heterogeneous social experiences and behaviors.

Michael Hanna, Ph.D.: And this is actually done in a mathematical precise way so there we have a hierarchical prompt.

Michael Hanna, Ph.D.: or partially observable mark of decision process model and we use it for trying to explain like social withdrawal or other social problems in depression.

Michael Hanna, Ph.D.: So it's a little bit rumination adjacent and it's more method sized pre print is available, but it's not yet through the revision process and what we also want to do is like.

Michael Hanna, Ph.D.: simulate distorted social functioning and mental disorders.

Michael Hanna, Ph.D.: And also to compare the simulations or computer simulations based on active inference to empirical data, because this is also what we found in the literature to be missing.

Michael Hanna, Ph.D.: We have the simulation studies right where we have like these simple games where active inference computer agents play.

Michael Hanna, Ph.D.: Like a trust game or something like that, or where you simulate hallucinations and psychosis and these are really good studies don't get me wrong, but what is missing in our view is a little bit combining these computer simulations with.

Michael Hanna, Ph.D.: The behavior of human being to compare them with on within one of the.

Michael Hanna, Ph.D.: One of the same design, and this is what we aim to do, but it's an ambitious project and we are not finished, so this will take more work.

Michael Hanna, Ph.D.: And another project that we actually will finish this year and we have done the literature screening is we have prepared like a systematic review.

Michael Hanna, Ph.D.: Of all active inference literature that is of interest in for psychological treatment and this will hopefully at least be a pre print this year.

Michael Hanna, Ph.D.: Yeah, so thank you, these are my co workers and different projects to drop a few names on the left is Matthias Feldman.

Michael Hanna, Ph.D.: Then Lucas Kirchner and Tobias Kube. These are the guys where that I wrote the rumination paper with.

Michael Hanna, Ph.D.: Then there's Anna Lena Eckert. She is from a theoretical cognitive science groups and she does a lot of the computer models for our new projects.

Michael Hanna, Ph.D.: So we need actually scientists that are better at programming than I am to and she's doing a lot of the work here and then there's Philip Herzog and Birgit Klein and with those two people.

Michael Hanna, Ph.D.: I also collaborated on a predictive processing model of PTSD, which was also.

Michael Hanna, Ph.D.: published in neuroscience and bio behavioral reviews at 2020.

Michael Hanna, Ph.D.: yeah so with that, I would like to thank you for listening and yeah we can discuss a few questions if you want now.

Michael Hanna, Ph.D.: awesome.

Michael Hanna, Ph.D.: Thank you.

Michael Hanna, Ph.D.: Okay.

Michael Hanna, Ph.D.: All right.

Michael Hanna, Ph.D.: Well, thanks for the great presentation.

Michael Hanna, Ph.D.: Looking forward to the discussion.

Michael Hanna, Ph.D.: So just to sort of begin.

Michael Hanna, Ph.D.: How did predictive processing and active become integrated with your and your colleagues research direction?

Michael Hanna, Ph.D.: Were you.

Michael Hanna, Ph.D.: Looking for integrative frameworks and came across this area or.

Michael Hanna, Ph.D.: Did this motivate you to study in enter into the clinical space.

Michael Hanna, Ph.D.: um.

Michael Hanna, Ph.D.: it's a mixture of two things I would say one is like observations and patients.

Michael Hanna, Ph.D.: Maybe start with that so.

Michael Hanna, Ph.D.: My friend Tobias Kube got me interested in this stuff and he had a patient and this patient was.

Michael Hanna, Ph.D.: Like really.

Michael Hanna, Ph.D.: had had to struggle with social phobia and.

Michael Hanna, Ph.D.: What was interesting about him that he was discarding information that would challenge his worldview so to say right so.

Michael Hanna, Ph.D.: Whenever somebody said that he actually liked him this one said all now he just pretends to say this because he wants to be a nice guy, but in reality, nobody likes me.

Michael Hanna, Ph.D.: And this is what got us interesting interested in this line of thinking about information seeking, for example, how do organisms seek information to.

Michael Hanna, Ph.D.: To run their lives, but also how do organisms discard information.

Michael Hanna, Ph.D.: And stay in a depressed or in an anxious state.

Michael Hanna, Ph.D.: And this is, I guess, one of the reasons why we started to get interested in active inference and another reason.

Michael Hanna, Ph.D.: comes from my love or love from a lot of my colleagues.

Michael Hanna, Ph.D.: For simple explanations and I don't mean simple as mathematically simple because I don't get most of the math, to be honest, fully but simple in the sense of we can have.

Michael Hanna, Ph.D.: Only a few axioms and explain sentient behavior with it, we can computer simulate behavior in it and it actually produces interesting behavior patterns that we know from the literature, and this is the other reason why we got into it.

Michael Hanna, Ph.D.: Awesome.

Michael Hanna, Ph.D.: So many directions and ways to go, but one excerpt from the paper that jumped out at me was the line between adaptive problem solving and self perpetuating maladaptive ruminating.

Michael Hanna, Ph.D.: So how do we know that line and how does that play out when you're in the clinical context?

Michael Hanna, Ph.D.: Like what is the appropriate amount of thinking or overthinking?

Michael Hanna, Ph.D.: How do we really find that line and come to know it?

Michael Hanna, Ph.D.: Well, there's a short answer that is we don't know.

Michael Hanna, Ph.D.: So it's hard to say, to be honest, where the line is, because I mean, in principle, maybe you find a good solution if you ruminate just two hours more, right?

Michael Hanna, Ph.D.: You never know, but we have an answer in the clinical field in practice and it's like a pragmaticist answer and it only goes okay if you suffer a lot and don't get your everyday life sorted out, then it becomes a problem.

Michael Hanna, Ph.D.: And we have like measures for that, so you can assess the global functioning level of a person and say, can he continue to work?

Michael Hanna, Ph.D.: Has he friends?

Michael Hanna, Ph.D.: Has he or she free time activities that actually fulfill her or him?

Michael Hanna, Ph.D.: And this way you can at least say okay.

Michael Hanna, Ph.D.: Has this person a good good level of functioning?

Michael Hanna, Ph.D.: And if not, does rumination impair this level of functioning?

Michael Hanna, Ph.D.: And yeah, this is a really pragmaticist answer, sorry for that, but this is how we do it in clinical practice.

Michael Hanna, Ph.D.: Totally, totally makes sense.

Michael Hanna, Ph.D.: Also in that kind of clinical area.

Michael Hanna, Ph.D.: So I was curious, other than rumination, what other transdiagnostic risk factors, what are some examples of those risk factors?

Michael Hanna, Ph.D.: And how might they be linked together in a kind of holistic understanding?

Michael Hanna, Ph.D.: Or what is the ACTIMF and non-ACTIMF way of thinking about the causes and consequences of these transdiagnostic risk factors?

Michael Hanna, Ph.D.: Yeah, yeah, one that jumps to mind is emotion regulation that comes up over and over again.

Michael Hanna, Ph.D.: So this is a risk factor also for depression.

Michael Hanna, Ph.D.: It's a risk factor for borderline personality disorder, but it's also a risk factor for violence.

Michael Hanna, Ph.D.: So a lot of things are connected to poor emotion regulation and I have to think shortly to get an active inference answer out of that.

Michael Hanna, Ph.D.: There's actually a paper, a new one by my colleague, Philipp Herzog, about like a predictive processing model of borderline personality disorder, where he describes in more detail.

Michael Hanna, Ph.D.: So he is the expert there.

Michael Hanna, Ph.D.: But.

Michael Hanna, Ph.D.: Maybe we can just look it up before I make stuff up that it's not too.

Michael Hanna, Ph.D.: Can I just search it shortly and give this an answer?

Michael Hanna, Ph.D.: Sure.

Michael Hanna, Ph.D.: This is a internet extended cognitive rumination.

Michael Hanna, Ph.D.: Yes, because I don't feel confident enough to talk about stuff that was not my main area of research.

Michael Hanna, Ph.D.: So.

Michael Hanna, Ph.D.: Extended rumination.

Michael Hanna, Ph.D.: Extended.

Michael Hanna, Ph.D.: You could make a script or a computer program that just iterates and whether it experiences that in an anxious way, kind of a second question, but at least from that functional side.

Michael Hanna, Ph.D.: It's it's as if ruminating.

Michael Hanna, Ph.D.: Yes.

Michael Hanna, Ph.D.: So I have the paper here.

Michael Hanna, Ph.D.: Do you see it now?

Michael Hanna, Ph.D.: Yes, thank you.

Michael Hanna, Ph.D.: Yeah.

Michael Hanna, Ph.D.: So let's see here.

Michael Hanna, Ph.D.: They actually have a model of active inference terms, so let's see where it starts.

Michael Hanna, Ph.D.: And it's actually not so simple.

Michael Hanna, Ph.D.: So okay, so what they want to explain is basically how childhood maltreatment leads to borderline personality disorder.

Michael Hanna, Ph.D.: And this is important for my transdiagnostic marker of reduced emotion regulation because bad emotion regulation is the symptom of borderline personality disorder.

Michael Hanna, Ph.D.: And what they say is okay, let's see that we have two relevant factors here.

Michael Hanna, Ph.D.: It's strong negative beliefs about the self and others.

Michael Hanna, Ph.D.: It's also typically known for borderline personality disorder.

Michael Hanna, Ph.D.: So we have high precision to use active inference terms here or Bayesian terms here.

Michael Hanna, Ph.D.: To my prior beliefs that other people are bad and I am also like a loser that cannot do anything right in his or her life.

Michael Hanna, Ph.D.: And this needs, of course, if the precision of the prior is really high, new information doesn't matter.

Michael Hanna, Ph.D.: And so we have a persistence of effects of hostility because if the others are all bad, well, I can be hostile towards them.

Michael Hanna, Ph.D.: Of fear and of sadness.

Michael Hanna, Ph.D.: And in addition to that also contributing to the temporal fluctuations because we know that borderline is characterized by sometimes.

Michael Hanna, Ph.D.: Like I really like my therapist as a borderline patients and after a while I really hate him and not a lot in between.

Michael Hanna, Ph.D.: And this may come about you to like a high uncertainty of beliefs about consequences of action so he has about policies so here my standard deviation is really broad and due to that.

Michael Hanna, Ph.D.: Like belief updating regarding policies gets fast, and this is why I show like instability, while at the same time also showing persistence and I show persistence of my worldview but instability of my behavior and.

Michael Hanna, Ph.D.: And this is how they try to explain borderline personality disorder in active inference terminology, but this is only me giving it a try I only read the paper once but this might be one way how to characterize.

Michael Hanna, Ph.D.: Like symptoms of emotional regulation problems in a conceptual model using Bayesian terminology.

Michael Hanna, Ph.D.: Awesome.

Michael Hanna, Ph.D.: And also in this paper and in yours it's on the roadmap towards formalization not like that's the only end point.

Michael Hanna, Ph.D.: But going from the empirical evidence and the ways of thinking and knowing in the clinician's office through the integrative frameworks to the pros and to the graphics that connect different nodes and edges.

Michael Hanna, Ph.D.: This figure and the ones that you drew, they're not exactly a Bayes graph, so it's not exactly ready to be entered into a computer program.

Michael Hanna, Ph.D.: Yes.

Michael Hanna, Ph.D.: But it's also not far from a Bayes graph.

Michael Hanna, Ph.D.: You have variables that could basically be variables in a Bayes graph.

Michael Hanna, Ph.D.: And so it moves along and brings collaborators and so on as in a kind of methodologically plural way there can be developments on multiple fronts and integrate the neuroimaging and the.

Michael Hanna, Ph.D.: Yeah.

Michael Hanna, Ph.D.: predispositions and then the treatment modalities in a way that's not just like following the causal chain of the etiology, but rather approaching it almost like in a reverse problem solving way starting with the clinicians awareness and view and first seeking to integrate and improve the sense making of the clinician even without a formal model.

Michael Hanna, Ph.D.: Yeah, I would totally agree to that not much I can say about it, I think it's totally right what you said there and maybe one thing.

Michael Hanna, Ph.D.: In clinical experience, at least in psychology I don't dare to speak of other fields, but they are backward translation is really important, so this is what you described right so going back from the clinical phenomenon to like more precise view, for example, in.

Michael Hanna, Ph.D.: In formal modeling and typically.

Michael Hanna, Ph.D.: Problems got tackled the other way around and this oftentimes doesn't work as well as backwards translation because typically problems got.

Michael Hanna, Ph.D.: For three by like.

Michael Hanna, Ph.D.: Basic scientists that are really good at doing this formal modeling but don't model necessarily the right parameters from our view, at least because they lack the clinical experience and so.

Michael Hanna, Ph.D.: Both parties working together is really important like guys like us.

Michael Hanna, Ph.D.: Bring in the important parameters or at least where we think they could be important from clinical practice and.

Michael Hanna, Ph.D.: The more mathematically inclined can model them and so there could be a fruitful collaboration between so start with backward translation then do forward translation then do another loop of backwards translation.

Michael Hanna, Ph.D.: And I guess this is a fruitful way of advancing a psychiatry and psychology.

Michael Hanna, Ph.D.: Wow.

Michael Hanna, Ph.D.: So.

Michael Hanna, Ph.D.: A related question.

Michael Hanna, Ph.D.: To what extent.

Michael Hanna, Ph.D.: Either today or in the future.

Michael Hanna, Ph.D.: Do you think patients.

Michael Hanna, Ph.D.: Would be clued in.

Michael Hanna, Ph.D.: To some of this lexicon and ontology.

Michael Hanna, Ph.D.: For example, is it enough to just say, hey, here's the new recommendation.

Here's the new exercise plan.

Michael Hanna, Ph.D.: Or is it explicitly oriented around communicating within these ways of talking about belief, uncertainty, information, and so on.

Ph.D.: I think it can be done with patients and it should be done to a certain extent.

Ph.D.: I think it's not plausible to teach them all the math and stuff like this will not be possible for at least a lot of them.

Michael Hanna, Ph.D.: Of course, sure, if you have a mathematics professor patient, it would be easy for him, but or her.

Michael Hanna, Ph.D.: But in most cases, no, not that, but you could talk about precision, you could talk about surprisal, you could talk about hierarchical learning.

Michael Hanna, Ph.D.: And I guess it's important to do this, you could simplify models into like vicious circles.

Michael Hanna, Ph.D.: One could easily draw a simple vicious circle with the help of our rumination model and teach them or the patients a few Bayesian.

Michael Hanna, Ph.D.: Adjacent words at least to better understand their condition.

Michael Hanna, Ph.D.: And I think this is important because psychoeducation actually is proven to help right the more you know about the condition.

Michael Hanna, Ph.D.: The higher your chances of getting better are in in a lot of cases, and so I guess yes, you could could teach them at least.

Michael Hanna, Ph.D.: base rules and a few simple things not doing a lot of math but.

Michael Hanna, Ph.D.: Like drawing a few graphs and try to teach them the basics would be a good idea, because I guess everybody.

Michael Hanna, Ph.D.: can get what precise information is and what.

Michael Hanna, Ph.D.: Like belief updating how it functions in principle, this is teachable to most of us.

Michael Hanna, Ph.D.: Yeah, one example that from the talk you mentioned, like with the loss of an important person in our life, our social or our emotional support policies are going to have to be updated.

Michael Hanna, Ph.D.: If there was some source we went to for information and that source no longer became available, we would have to change our epistemic foraging.

Michael Hanna, Ph.D.: And of course, this exists in the social space.

Michael Hanna, Ph.D.: And so when life changes, when our generative model or when our generative process, our niche changes, we should expect the unexpected.

Michael Hanna, Ph.D.: And so it's a totally normal, surprising, but expected surprise event that opens up the discussion to like, there will be beliefs that will be changing.

Michael Hanna, Ph.D.: There will be new policies or policies that you didn't previously take.

Michael Hanna, Ph.D.: Maybe now's the time to try them out and there's going to be volatility as you're learning the consequences of action.

Michael Hanna, Ph.D.: Yes.

Michael Hanna, Ph.D.: So stick with it and try something three times before saying that it doesn't work even without appeal to mathematics.

Michael Hanna, Ph.D.: That's very powerful.

Michael Hanna, Ph.D.: And I think the way that somebody might say, well, in this double blind study, these two groups, this treatment that we're going to go with, it's been shown that in this context, these two groups, this group did better.

Michael Hanna, Ph.D.: And no one says, well, slow down, you know, the T-test used a lot of mathematics.

Michael Hanna, Ph.D.: I'm not a statistics professor.

Michael Hanna, Ph.D.: So people accept the epistemology of the group differential statistics and of diagnostic statistics, even though one could do 100 PhDs in theoretical statistics and in computational statistics.

Michael Hanna, Ph.D.: And so it just speaks, I think, to the early stage when the field is even perceived as being math heavy because we're in such a methodological developmental stage.

Michael Hanna, Ph.D.: Yes.

Michael Hanna, Ph.D.: But there will be layers and ways of working that will not mention any more technical detail than someone just saying the double blind study has provided evidence for X.

Michael Hanna, Ph.D.: Yes, yeah, yeah, I can say two things about it.

Michael Hanna, Ph.D.: The one thing is a philosophical thing.

Michael Hanna, Ph.D.: Daniel's net talked about free floating rationales right so a free floating rationale in case you don't know is like.

Michael Hanna, Ph.D.: When lions go for gazelles it's the example that he likes to give the gazelles to these show boating types of jumps.

Michael Hanna, Ph.D.: And the lions typically don't go after the gazelles that can show this behavior, but go after the injured ones that cannot do this anymore.

Michael Hanna, Ph.D.: Right, but we can be quite certain that neither the lion nor the gazelle knows what they are doing, but it works without knowledge.

Michael Hanna, Ph.D.: And this is what he calls a free floating rationale.

Michael Hanna, Ph.D.: There is a rationale behind the behavior of the lion, but he doesn't need to know it in order to take advantage of it.

Michael Hanna, Ph.D.: And this is the case for science, and this is also the case for patients you don't need to know every detail of a mathematical formalism in order to grasp some of the concepts and.

Michael Hanna, Ph.D.: To advance you.

Michael Hanna, Ph.D.: Your understanding of the world using these concepts.

Michael Hanna, Ph.D.: But it's of course it helps.

Michael Hanna, Ph.D.: But.

Michael Hanna, Ph.D.: For patients it's the same and for us it's the same and.

Michael Hanna, Ph.D.: And I think.

Michael Hanna, Ph.D.: Free floating rationales are really important concept to describe why a lot of the times you don't need to know the exact formalisms.

Michael Hanna, Ph.D.: And another idea that I want to give is more like of science.

Michael Hanna, Ph.D.: Of making niche constructions more applicable to the mainstream.

Michael Hanna, Ph.D.: And this is.

Michael Hanna, Ph.D.: Well, at the beginning it's hard because you need to program every little bit yourself.

Michael Hanna, Ph.D.: There are no toolboxes nothing to help you and not really books just a few experts,

but after a while books come out.

Michael Hanna, Ph.D.: After a while you get toolboxes that actually simplify a lot of the process you don't.

Michael Hanna, Ph.D.: You don't need to write the helper functions yourself, you can use the toolbox in a lot of cases and so after a while it becomes more widely applicable in science and maybe at Sunday.

Michael Hanna, Ph.D.: I don't know SPSS or something one of these statistics software's maybe not has like a way to click through active inference algorithms let's see.

Michael Hanna, Ph.D.: But then, in principle, everybody can use it when you have new problems because people will use it without actually understanding it, but.

Michael Hanna, Ph.D.: I would predict that it gets simpler and simpler to to use and I would say.

Michael Hanna, Ph.D.: Overall, this is a good thing, because this allows a lot more scientists to jump in on the endeavor.

Michael Hanna, Ph.D.: Thanks yeah great points.

Michael Hanna, Ph.D.: One of your quotes the short one was existence precedes essence.

Michael Hanna, Ph.D.: And it reminded me of a 2015 paper of fristen and mathis, which is entitled I am therefore I think.

Michael Hanna, Ph.D.: And that of course is a little bit of an inversion on I think therefore I am, which puts rumination before the horse or the cart before the horse or I'm not exactly sure.

Michael Hanna, Ph.D.: But I think therefore I am.

Michael Hanna, Ph.D.: It actually puts thinking and ruminating first, whereas this existentialist quote that you provided concords well with certain ways of thinking about reflexive systems in active inference.

Michael Hanna, Ph.D.: So what did you mean with the introduction of that quote and how does that existentialist angle play out in the clinical setting?

Michael Hanna, Ph.D.: Yeah, these are maybe two questions disguised as one.

Michael Hanna, Ph.D.: I start with the existentialist worldview in clinical settings.

Michael Hanna, Ph.D.: Well, I think it's important for clinical science because and more so for clinical practice because.

Michael Hanna, Ph.D.: We need to believe in a certain type of liberty in order to help patients if we believe everything's predetermined by I don't know.

Michael Hanna, Ph.D.: Brain chemistry, even if it's true.

Michael Hanna, Ph.D.: This will make changing your behavior hard because you can always say, well, I cannot change my behavior.

Michael Hanna, Ph.D.: It's all in my I don't know genes or what type of biological reductionism you like.

Michael Hanna, Ph.D.: So I guess we need these existentialist idea that you yourself are in control of your life in order to make therapy work.

Michael Hanna, Ph.D.: Right.

Michael Hanna, Ph.D.: So we need to at least have a compatible.

Michael Hanna, Ph.D.: Hard English words like this.

Michael Hanna, Ph.D.: Even if determinism is true, we need to have.

Michael Hanna, Ph.D.: It's combined with some sort of liberty or belief in liberty, and I guess this is where active inference jumps in because active inference this allows a formal description about self control and then

we have a definition of.

Michael Hanna, Ph.D.: Of.

Michael Hanna, Ph.D.: Of.

Michael Hanna, Ph.D.: Of.

Michael Hanna, Ph.D.: Of acting in the world and learning new things that doesn't require like fairy tales and it's compatible with determinism and with this world for you, you have a world view that is.

Michael Hanna, Ph.D.: On one hand is compatible with determinism and on the other hand allows for a concept like self control agency policy learning all the things, and so it can also be used for psychotherapy.

Michael Hanna, Ph.D.: Yeah.

Michael Hanna, Ph.D.: Interesting.

Michael Hanna, Ph.D.: I hope this answers the question, but this is what I can say.

Michael Hanna, Ph.D.: Yeah, there's so much depth, obviously, many threads and traditions in the clinical setting and that full stack from the manifesto,

Michael Hanna, Ph.D.: The manifest behavioral on through the cognitive and psychological, but verging or anchored in or descended from the metaphysical.

Michael Hanna, Ph.D.: That's that's a full spectrum area that seems like it must be navigated in the clinician's education, but even in like the one hour that you're spending with somebody, I imagine that it could probably oscillate a lot in where you're focusing from here's how to tie your shoe to here's why things matter.

Michael Hanna, Ph.D.: Yeah.

Michael Hanna, Ph.D.: Yeah.

Michael Hanna, Ph.D.: Yeah.

Michael Hanna, Ph.D.: Yeah.

Michael Hanna, Ph.D.: There's more philosophy, especially practical philosophy for therapy than one would think.

Michael Hanna, Ph.D.: Because patients bring it up all the time, right?

Michael Hanna, Ph.D.: I mean, one of the problems of depression, if it's going to be really severe, this, for example, nihilism kicks in.

Michael Hanna, Ph.D.: And so nothing makes sense anymore.

Michael Hanna, Ph.D.: And then, as you said, why should anything matter at all then becomes an interesting question and depressing question for them.

Michael Hanna, Ph.D.: What's your go to response?

Michael Hanna, Ph.D.: Why does anything matter?

Michael Hanna, Ph.D.: Oh, it's a hard question.

Michael Hanna, Ph.D.: I would try to get, do you always thought that nothing matters?

Michael Hanna, Ph.D.: What was different when something mattered for you?

Michael Hanna, Ph.D.: How was your life then?

Michael Hanna, Ph.D.: And what changed?

Michael Hanna, Ph.D.: Why you came to these conclusions now?

Michael Hanna, Ph.D.: And how do you feel when you think about your old worldview versus new worldview?

Michael Hanna, Ph.D.: So I would try to challenge their worldview by their own worldview a few years ago.

Michael Hanna, Ph.D.: This is not always working because you have patients that have actually gone through a lot of difficult times in their lives, but typically like their worldview was not always depressed.

Michael Hanna, Ph.D.: Yes, that seems to do multiple things by responding to a question with a question.

Michael Hanna, Ph.D.: It puts an active posture back into the conversation.

Michael Hanna, Ph.D.: So at the very least policies can be selected that aren't rumination and consequences could be learned, even if there's going to be a higher low precision on that learning.

Michael Hanna, Ph.D.: At least it like opens the door to movements, whereas just well person X said this or that that response may land or inspire or not, but it reifies the pacification.

Michael Hanna, Ph.D.: And therefore may not at kind of like an inactive layer promote the movement.

Michael Hanna, Ph.D.: Yeah.

Michael Hanna, Ph.D.: Yeah.

Michael Hanna, Ph.D.: Yeah.

Michael Hanna, Ph.D.: Yeah.

Michael Hanna, Ph.D.: I just wanted to say, yes.

Michael Hanna, Ph.D.: Oh, awesome.

Michael Hanna, Ph.D.: I think the same.

Michael Hanna, Ph.D.: And if you try something like person XY that says, this is the reason why your life works.

Michael Hanna, Ph.D.: This will in most cases not work at all, they will say something like, oh, but this person doesn't go through that many hardship that I did, what is oftentimes true.

Michael Hanna, Ph.D.: Or something like that, right?

Michael Hanna, Ph.D.: So often it's better if the ideas are sampled from their own generative models from a lost part of the distribution, so to say, and not from outside.

Michael Hanna, Ph.D.: Typically, you have to try to get them to think out of their own worldview and not of your worldview or someone else's worldview.

Michael Hanna, Ph.D.: This seldom functions.

Michael Hanna, Ph.D.: Hmm.

Michael Hanna, Ph.D.: So another interesting area, you mentioned the problem, difficulty, complexity, time, uncertainty, estimation.

Michael Hanna, Ph.D.: So kind of like a cognitive task analysis in the CBT setting where different tasks are estimated, again, along these axes of how challenging is that for you or how complex does that task seem to you?

Michael Hanna, Ph.D.: Yeah.

Michael Hanna, Ph.D.: How does that really play out?

Michael Hanna, Ph.D.: Because it really seems like something in the email inbox or in writing a paper with some improved signposting or learning, even, around which tasks are hard, which tasks are clear, which tasks are threatening,

Michael Hanna, Ph.D.: threatening our wellbeing, those kinds of metadata on policy selection and being able to see that could help us select better policy and not to change our preferences, which is how sometimes

behavioral change is approached by changing what people are supposed to want.

Michael Hanna, Ph.D.: But by bringing this kind of cybernetic clarity around action selection to support people in different contexts, making the decisions that can then credibly be said to be more in alignment with what they do want.

Michael Hanna, Ph.D.: Yeah.

Michael Hanna, Ph.D.: Yeah, I mean, what is typically already been done is to let patients guess how high they would rate the probability of succeeding, right?

Michael Hanna, Ph.D.: So when we do this behavioral therapy, they are typically asked, okay, what do you think?

Michael Hanna, Ph.D.: How probable is it that you're going to go there until next week?

Michael Hanna, Ph.D.: And it's a rule of thumb by clinicians.

Michael Hanna, Ph.D.: It has no base in science, but we say if it's 80% or higher probability, one should aim for it.

Michael Hanna, Ph.D.: If it's lower, it's too difficult.

Michael Hanna, Ph.D.: But if it's trivial, if it's 99.9% probability, it's also not worth typically pursuing, at least not always.

Michael Hanna, Ph.D.: But I guess a problem with that is that I don't know a solution for this is my subjective probability right how I think how probability is I don't have a good grasp in all cases on how difficult it really is.

Michael Hanna, Ph.D.: For me, because there are all sorts of biases that would keep me from making such estimations.

Michael Hanna, Ph.D.: I don't have a solution for that, but what would be good is to have.

Michael Hanna, Ph.D.: More variables that we could estimate in order to actually have more information on how difficult a task is, but for most cases we haven't.

Michael Hanna, Ph.D.: And if when we have them, we know that what patients think how difficult it is and how difficult it really is hasn't the best correlation sure it's positively correlated.

Michael Hanna, Ph.D.: But how good I think my short term memory, for example, is and how good it really is is not the same and this is a hard to solve problem for clinical science.

Michael Hanna, Ph.D.: I have to think about it longer, but it would.

Michael Hanna, Ph.D.: Giving patients to it to actually improve their probability estimates would be great.

Michael Hanna, Ph.D.: But I guess it's also hard to do due to all these things that you said before right if the problem is complex multi layered and has a lot of areas and good luck actually finding.

Michael Hanna, Ph.D.: A way to estimating probabilities, but we could start for simple problems and maybe this helps patients, but I have to think about it more to be honest.

Michael Hanna, Ph.D.: It's not easy to find better solutions there.

Michael Hanna, Ph.D.: Hmm.

Michael Hanna, Ph.D.: Yeah, even just on paper or digitally journaling, but then eventually potentially integrating with the kinds of formal models we've been discussing.

Michael Hanna, Ph.D.: It could be really nice to know.

Michael Hanna, Ph.D.: Hey, there's a lot of stuff that you said had a below 50 chance of happening and nine out of 10 times you've done that.

Michael Hanna, Ph.D.: So maybe next time you're in the zone of uncertainty, just go for it.

Michael Hanna, Ph.D.: Yeah.

Michael Hanna, Ph.D.: Yeah.

Michael Hanna, Ph.D.: And or just this path, you tend to systematically overestimate your confidence in this or you over underestimate how much other people will appreciate this kind of consequence.

Michael Hanna, Ph.D.: And maybe writing three thank you notes a day.

Michael Hanna, Ph.D.: Yeah.

Michael Hanna, Ph.D.: Would be a great use of 20 minutes.

Michael Hanna, Ph.D.: Yeah, these are all really good ideas.

Michael Hanna, Ph.D.: And it's, this is also things that are done right what what is missing, but what is doable in principle is like tracking certain behaviors like.

Michael Hanna, Ph.D.: And an easy example would be aggressive behavior, for example, counting how much like verbal aggression was shown or physical aggression was shown over the days and then like.

Michael Hanna, Ph.D.: Look what the expectation of the patient was and comparing those two and they could do like basic belief updating in a simplified form.

Michael Hanna, Ph.D.: Yeah, yeah, this would be a great idea.

Michael Hanna, Ph.D.: It's always then possible, I would say, when the behavior is clear enough and easy enough to count right as soon as if it's getting more unclear.

Michael Hanna, Ph.D.: This will not work, but it could work for a lot of problem behaviors like self harm aggression.

Michael Hanna, Ph.D.: Not lying in bed like for X in depression, something like that.

Michael Hanna, Ph.D.: So it makes me think about that contrast.

Michael Hanna, Ph.D.: That you provided between reward oriented frameworks and which would struggle to describe rumination as it's not in the short or medium or long term necessarily oriented with rewards.

Michael Hanna, Ph.D.: So the only explanation becomes, well, something is off in the reward function, which is kind of tautological.

Michael Hanna, Ph.D.: And so that's a little bit of a local and a constrained explanation.

Michael Hanna, Ph.D.: Whereas when we see policy selection, external overt movement in the world, as well as internal covert policy selection, like attention and mental action.

Michael Hanna, Ph.D.: When we see those actions as being oriented around reduction of free energy, which is bounding surprise and therefore based upon what is likely, not what is necessarily the most rewarding.

Michael Hanna, Ph.D.: It opens up that view to again say, how many of this behavior do you expect to happen in the next week?

Michael Hanna, Ph.D.: Yes, in a dream world, we would aim for zero of events of this time.

Michael Hanna, Ph.D.: You know, we'll aim for you to be full time employed by next week.

Michael Hanna, Ph.D.: That sounds awesome.

Michael Hanna, Ph.D.: But what do we honestly expect in one, two, three, four, five weeks?

Michael Hanna, Ph.D.: And then we're going to actually be looking back in and then it's instead of maybe feeling like, well, you're going to leave the port and then you're going to have to sail across an ocean.

Michael Hanna, Ph.D.: It can be more like there are checkpoints.

Michael Hanna, Ph.D.: We're looking back at the buoys that we passed.

Michael Hanna, Ph.D.: We have the next three buoys that we're looking towards.

Michael Hanna, Ph.D.: We can do smaller corrections and there will be volatility.

Michael Hanna, Ph.D.: So there's going to continue to be surprise.

Michael Hanna, Ph.D.: We're not going to exactly hit every milestone per se, but if we are on this trajectory, we're going to be moving essentially as much as we possibly could in the right direction.

Michael Hanna, Ph.D.: Yes.

Michael Hanna, Ph.D.: And spending less time just hypothesizing about like a total counterfactual world where the issue is already resolved.

Michael Hanna, Ph.D.: Yes.

Michael Hanna, Ph.D.: And this is a metaphor metaphor that is also used in therapy quite often, right?

Michael Hanna, Ph.D.: The one of the mountain climber and the mountain climber typically needs a guide if he's a newbie.

Michael Hanna, Ph.D.: And what psychotherapy can do is like provide like a few a good way over the mountain, so to say, provide a good place for.

Michael Hanna, Ph.D.: For resting doing the next step, doing the next step, doing the next step.

Michael Hanna, Ph.D.: And yeah, this is really resonating with what you said, namely, we have to.

Michael Hanna, Ph.D.: Take a big problem and cut it into little problems that we can solve one after another so.

Michael Hanna, Ph.D.: It becomes plausible to actually solve it.

Michael Hanna, Ph.D.: Yes, I would totally agree.

Michael Hanna, Ph.D.: Yeah.

Michael Hanna, Ph.D.: Cool.

Michael Hanna, Ph.D.: And the emotional flanker tasks.

Michael Hanna, Ph.D.: I'm not exactly sure what those would be, but it just made me think of our policy selection throughout the day.

Michael Hanna, Ph.D.: And all of the.

Michael Hanna, Ph.D.: Fires we have to put out or all of the bubbling soups that we have to tend to with thirst and rest and breathing and physical exercise and all of the

Michael Hanna, Ph.D.: All of these different aspects to our cognitive and emotional wellbeing.

Michael Hanna, Ph.D.: It's like there's many things to attend to, which is why our attention is so variable as a feature.

Michael Hanna, Ph.D.: It's not a bug.

Michael Hanna, Ph.D.: So the fact that we have.

Michael Hanna, Ph.D.: Exactly.

Michael Hanna, Ph.D.: Pathological or not just neurodiverse states that involve a fixation of attention and also the inability to fixate attention.

Michael Hanna, Ph.D.: It just starts with.

Michael Hanna, Ph.D.: Our experience of the day and of our attention to our body and our surroundings and

to the bigger questions that we know impinge on us, but also we can't necessarily.

Michael Hanna, Ph.D.: Have the geopolitics and the climate change.

Michael Hanna, Ph.D.: Solved just by thinking about it.

Michael Hanna, Ph.D.: But we also don't want to neglect what are important issues and then seeing strategies and having the discussion about strategies and tactics in that attentional space instead of in the reward space.

Michael Hanna, Ph.D.: Not to bash or anything like that, but that conversation is about, well, what should we value more or less?

Michael Hanna, Ph.D.: And then why are these values translating effectively or not into behavior?

Michael Hanna, Ph.D.: Yeah.

Michael Hanna, Ph.D.: Whereas the attention and surprise angle is like being more in the game and your local view instead of like watching the game and just saying, well, I want this to be the outcome.

Michael Hanna, Ph.D.: Exactly.

Michael Hanna, Ph.D.: Yeah.

Michael Hanna, Ph.D.: I want to say a little bit about this attention focus and attention spreading that you mentioned.

Michael Hanna, Ph.D.: I almost said it in the talk, but I say it now.

Michael Hanna, Ph.D.: Like, for example, ADHD is not necessarily like a disorder in the sense that it's a dysfunction, right?

Michael Hanna, Ph.D.: So it can also be described or maybe better described as a neurodivergent property of some human beings.

Michael Hanna, Ph.D.: So they shift attention more easily.

Michael Hanna, Ph.D.: And this is depending on the environment, a good thing, because if you stuck your attention at one place too much, well, you might not notice what is going around in your surrounding.

Michael Hanna, Ph.D.: So you have to be able to focus away from certain points of attention.

Michael Hanna, Ph.D.: And you want to have in a group individuals that have really sticky attentions so they can focus on a task really good, but you also want to have the other ones.

Michael Hanna, Ph.D.: That have like really floating attention spans in order to be new opportunities or see danger.

Michael Hanna, Ph.D.: And so I think this can be really easily explained by evolutionary theory, right?

Michael Hanna, Ph.D.: So we have variants over the population and this variance is not a bug.

Michael Hanna, Ph.D.: It's a feature and on the more extreme ends of the variance, you find.

Michael Hanna, Ph.D.: Phenotypes that are problematic from time to time, but depending on the surroundings always depending on the surroundings right?

right so on the one hand you might have something like ocd like behavior and on the other hand something like more adhd like behavior and it can even jump between the two interestingly but this is not necessarily always bad it really depends on your environment and that humans differ with respect to this parameters is also not a bug but a feature of evolution i would say and also a feature of our learning history of course

totally agreed for individual and collective behavioral strategies they're often discussed as if they were context independence when there are ant colony foraging strategies that work in the desert and they work not as well in the jungle so just like it is with our attention there's attentional surroundings that support this kind of work but not that kind of work or well-being so um being able to talk about the generative model and the generative process and their engagement in a rich way

i guess it's what brings us together but it's also what feels like a uh powerful place to re-gather and remember a lot of the important qualitative threads and empirical threads in this important clinical area and also move forward in a way that's not just adding like another uh

measurement onto the pile because a lot of these models as you continue to develop their formality and ability to integrate data they will provide a new view on previous findings and patterns that may have been taken uh in the t-test reward learning world to be conclusive or this or that will be shown to be just two darts thrown into a huge neurodiversity space and there might be other studies or sets of studies where what was treated as being all over the place

are now like perfectly in line and tell a really coherent story about

for example how strong negative beliefs about the self and others or high uncertainty of beliefs about the consequences of action

end up helping us just do what we do

yes yeah totally agree it's uh maybe it's interesting there's even reconceptualization going on like uh thinking about psychopathology in more like evolutionary terms like uh stefan hofman for example and steven hayes

steven hayes is known as the founding father of um act so acceptance and commitment based therapy and those two guys have a new therapy idea where we describe um psychopathology on a pragmatic basis basis so we

define psychopathology as behavior that doesn't fit any more the current circumstances so there you have the context in right

right and i think it's clever to do it more that way because for example ptsd like behavior well if you're a soldier in a

in a really bad region of the world this is not psychopathology this is allowing you to survive

the problem is the problem is of pthd that the war comes home with you or traumatic event comes home with you

and now your behavior doesn't fit the environment anymore right and i think this is a neat way of describing a lot of

psychopathology as maladaptive behavior in the evolutionary sense and typically it was adaptive in the past

um like also what this uh like borderline personality disorder slide that i have on the screen right now

it's the same because it was adaptive uh in the past to behave like this um in in some circumstances

in a in a bad childhood so to say but it's not anymore but it's continued despite being ineffective

and i guess this is a good definition for a lot of psychopathology not for all i i would say for

example there is really psychopathology where you have broken inference so to say and not just um for example in in psychosis in some cases i would say yes uh you have more like brain pathology you have more really broken inference and the rules of active inference don't work anymore so to say but i don't think that the rules of active inference don't work in rumination they work perfectly fine but um the organism has placed themselves in uh in a situation where they don't learn anything new by applying the same rules that everybody else applies so it's broken models not broken inference and i guess this is true for a lot of psychopathology

yeah um so what are your next moves with uh this line of research you mentioned some of the papers what would you like those papers to show or who would you like them to speak to and show what um maybe i start with the literature uh review that we are planning i would like to get to know what the the literature so to say says about active inference informed treatments this is what i would like to find out i don't have a lot of hypotheses i want to screen the literature and find out whether we can synthesize them into an active inference informed model of what is psychotherapy doing and i would like to speak this model to a clinical audience in the first place again so i want uh the clinical population to learn more about active inference i know that psychiatrists sometimes know about active inference literature but psychologists at least not clinical psychologists typically don't know a lot about it and i would like to change this a little bit at least and afterwards um the next logical step would be to implement some of the insight into actual therapy programs but this is something for more of the future but it would be interesting whether you can really have something like an active inference informed therapy but now we are talking more like five to ten years

awesome do you have any final thoughts or questions and then i will close with a short quote to fire back yes you are sure no i don't have any final remarks or questions i would like to thank you for giving me the opportunity to speak and now i want to resolve my uncertainty regarding the quote

great the quote is from henry david thoreau and it's drawing a connection between the paths that we ruminate on that we walk upon and travel upon and then the paths in the mind so i thought it was so throw wrote i had not lived there a week before my feet wore a path from my door to the side and though it is five or six years since i trot it it is still quite distinct it is true i

fear that others may have fallen into it and so help to keep it open the surface of the earth is soft and impressible by the feet of men and so with the paths which the mind travels how worn and dusty then must be the highways of the world how deep the ruts of tradition and conformity

thank you awesome thank you for having me till next time thanks a lot see you

you